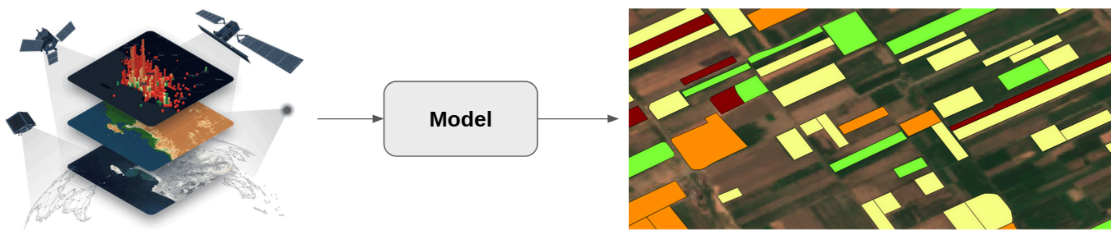
University of Science and Technology Houari Boumediene
Faculty of computer science

M1 SII 2025/2026

Project: Deep Learning for Crop Classification Using Multi-Source Satellite Data

Monitoring crop types (wheat, rice, soybean, …) over large areas is essential for agricultural management, food security assessment, and environmental monitoring. Satellite imagery provides a powerful way to map crops at large scales, especially when using time-series observations that capture the phenological development of vegetation throughout the growing season. Recent advances in deep learning have significantly improved crop classification accuracy, particularly, hybrid architectures combining convolutional neural networks (CNNs) and Transformers have shown strong performance in modeling temporal dependencies in satellite data.



Satellite Images Crop Prediction

However, crop classification accuracy can potentially be improved by integrating additional environmental information such as climate variables, soil properties, and topographic features. In this project you will investigate how such covariates influence crop classification performance and explore the design of improved models for multi-source agricultural data through 3 main steps:

<u>Part 1:</u> Base Classification Model

The goal of this first part is to reproduce the methodology described in the paper: **"***A lightweight CNN-Transformer network for pixel-based crop mapping using time-series Sentinel-2 imagery.***"**. You must replicate the data preparation pipeline and implement the proposed architecture to perform crop classification.

1- Literature Review: Read and analyze the methodology presented in the paper, focusing on: Data preparation, Temporal sampling strategy, Model architecture, Training procedure, and Evaluation metrics.

2. Dataset Acquisition: Download the datasets (Sentinel-2 & Cropland Data Layer Data) used in the paper for the two study areas in the United States (California & Arkansas).

3. Data Exploration: Perform exploratory analysis including: visualization of time-series vegetation indices, class distribution analysis, temporal patterns of different crops, and inspection of missing values or noise.

4. Data Preprocessing: Apply the preprocessing pipeline described in the paper, including: cloud filtering, time-series interpolation, normalization, extraction of vegetation indices, alignment with crop labels, etc.

5. Model Implementation: Re-implement the architecture proposed in the paper, train the model on the provided datasets, evaluate the classification results, and compare them with those reported in the paper.

Part 2: Integration of Environmental Covariates

The objective of this part is to evaluate how additional environmental variables affect crop classification performance. You will extend the dataset by incorporating environmental covariates related to the study area and conduct an ablation study.

1. Data Preparation: Download relevant covariate datasets (Climate, Soil, Topography), reproject and resample them to match the baseline dataset resolution, and preprocess.
2. Exploratory Analysis: Perform an exploratory data analysis on the new datasets separately and study the correlation between covariates and crop classes.
3. Ablation Study: Train the previously implemented model under different configurations:

1. Sentinel-2 data only (baseline)
2. Sentinel-2 + climate variables
3. Sentinel-2 + soil variables
4. Sentinel-2 + topography
5. Sentinel-2 + all covariates combined

Compare the performance of each configuration and analyse the results.

Part 3: Model Design and Improvement

In the final part, you must design and evaluate your own model architecture inspired by the scientific literature. You should consult the additional research papers furnished, or other ones you may find interesting.

1. Literature Review: Review additional relevant papers on crop mapping or satellite time-series modeling and Identify limitations of the reproduced model.
2. Proposed Model: propose an improved architecture, the design must be clearly justified. Train and evaluate the proposed model on the prepared dataset and evaluate its performance using the same metrics used in Part 1. Compare results with the covariate-enhanced model from Part 2.

Final Deliverable

You must produce a technical report describing the entire project. It should include:

1. Introduction and problem description
2. Description of datasets & preprocessing pipeline
3. Reproduction of the baseline model
4. Covariate integration and ablation study
5. Proposed model architecture
6. Experimental results, evaluation, and Discussion
7. Conclusion and possible future improvements

The report should also include:

- Task repartition of the team members
- References
- No more than 20 pages